# Fin 2020 Final Questions

- Student Name:
- Student ID:
- Class time:

Instructions:

- This exam has 7 questions. Including blank pages, there are 14 pages in total. You have 120 minutes to finish the exam. Late submission will not be accepted.

- *Provide concise proof of all answers.* If you show the steps that led to your results, I can award *partial credit* for the correct approach even if your final answers are slightly off.

**Question 1 [8 points]**
We are in the world of CAPM model. The market portfolio has an expected return of 15% and a standard deviation of 0.3. The risk free rate is 5%.

**Q1. A [4 points]**
Now you know a stock S has standard deviation of 0.3 and beta of 1.5. What should be the expected return of stock S?

**Q1. B [4 points]** You also know an efficient portfolio P (i.e. a portfolio on the capital market line, or equivalently, a portfolio formed out of risk-free asset and the tangency portfolio that consists of all the risky assets), the portfolio P has a standard deviation of 0.1. What should be the expected return of this portfolio P?

**Question 2 [10 points]**

True or False or "it depends". Briefly explain or qualify your answer.

**Q2. A [3 points]**

CAPM says that all risky assets must have positive risk premium, that is, all risky assets must have higher expected returns than risk free asset.

**Q2. B [4 points]**

CAPM assumes that all investors have the same information.

CAPM also says that market portfolio is the best portfolio in terms of risk return trade off when forming a portfolio that only consists of risky assets. That is, for any other portfolio that consists of only risky assets, if the portfolio has the same standard deviation as the market portfolio, it must have smaller or equal expected returns.

**Q2. C [3 points]**

APT says that no risk can be diversified away, even for well diversified portfolios

**Q3 [7 points]**

In class, we show that asset pricing models can be written as $1=E(mR_i)$, where m is a stochastic discount factor and $R_i$ is the gross returns for any asset i, including risk free asset, market portfolios. What differ one model from another is that m takes different forms. Just as how we did in class, show that when the stochastic discount factor, m, take the form of market returns or market returns+ some constant, you can derive the formula of CAPM model.

**Q4 [20 points]**

Consider an economy in which the random return $\tilde{r}_i$ on each individual asset i is given by

$$\tilde{r}_i = E(\tilde{r}_i) + \beta_{i,m}[\tilde{r}_M - E(\tilde{r}_M)] + \beta_{i,S}[\tilde{r}_S - E(\tilde{r}_S)] + \beta_{i,H}[\tilde{r}_H - E(\tilde{r}_H)] + \varepsilon_i$$

Where, similar as we discussed in class, $E(\tilde{r}_i)$ equals the expected returns on asset i, $\tilde{r}_M$ is the random return on the market portfolio, $\tilde{r}_S$ is the random returns on a portfolio of small firms. $\varepsilon_i$ is an idiosyncratic, firm specific component, and $\beta_{i,m}$, $\beta_{i,s}$ , $\beta_{i,H}$are the "factor loadings" that measure the extent to which the return on asset i is correlated with the return on the market portfolio, the portfolio of small size firms, and the portfolio of high book-market ratio firms. Assume, as Arbitrage Pricing Theory (APT) in class, that there are enough individual assets for investors to form many well-diversified portfolios, and that investors act to eliminate all arbitrage opportunities that may arise across all well-diversified portfolios.

**Q4.A [10 points]**

Suppose you find a well-diversified portfolio that has non-zero values of both $\beta_{w,m}$ and $\beta_{w,s}$, and that has expected returns

$$E(\tilde{r}_w) = (1 - \beta_{w,m} - \beta_{w,H})\, r_f + \beta_{w,m} E(\tilde{r}_M) + \beta_{w,s} E(\tilde{r}_S) + \beta_{w,H} E(\tilde{r}_H) + \Delta$$

Where $\Delta > 0$. Explain how you can use this portfolio and some well diversified portfolios you construct yourself to arbitrage.

**Q4.B [10 points]**

Now suppose that the return of asset i taking the form

$$\tilde{r}_i = E(\tilde{r}_i) + \beta_{i,m}[\tilde{r}_M - E(\tilde{r}_M)] + \beta_{i,SMB}[\tilde{r}_{SMB} - E(\tilde{r}_S)] + \beta_{i,HML}[\tilde{r}_{HML} - E(\tilde{r}_{HML})] + \beta_{i,Mom}[\tilde{r}_{MoM} - E(\tilde{r}_{\tilde{r}_{MoM}})] + \varepsilon_i$$

where $\tilde{r}_{SMB}$ is the return from the *long-short* portfolio that takes a long position in the portfolio of small stocks and takes a short position in the portfolio of large stocks, $\tilde{r}_{HML}$ is the return from the *long-short* portfolio that takes a long position in the portfolio of high book-to-market ratio stocks and takes a short position in the portfolio of low book-to-market ratio stocks, $\tilde{r}_{MoM}$ is the return from the *long-short* portfolio that takes a long position in the portfolio of high momentum stocks (i.e. stocks with high returns in the past year) and takes a short position in the portfolio of low momentum stocks (i.e. stocks with low returns in the past year). To avoid arbitrage, for a well-diversified portfolio with non-zero values of factor loadings $\beta_{w,m}$, $\beta_{w,s}$, $\beta_{w,HML}$ and $\beta_{w,Mom}$, what should be expected returns of this portfolio? Please show how you arrive at your conclusion.

**Question 5 [10 points]**

In Arrow-Debreu model, the price of state contingent claim for a state i, $q^i=\beta\pi_i u_j'(c_j^i)/u_j'(c_j^0)$ where $\pi_i$ is the probability of state i, $c_j^i$ is the consumption of agent j in state i, and $c_j^0$ is the consumption of agent j in current period.

Q5.A **[5 points]** What would be the expected returns of holding a state contingent claim i?

Q5.B **[5 points]** How should expected returns vary with the consumption in the state? What's the economic intuition behind the results?

**Question 6 [35 points]**

Consider a one good, 2 periods (t=1, 2) Arrow Debreu economy with 1/2 of investors type A, and 1/2 of investors type B. There are 2 states of the world indexed by state s=1, 2, the probability of state 1 is 3/4 and the probability of state 2 is 1/4. The endowments for agent A are $e^A = (1, (2, 4))$ (3 units of endowment at t=0, 2 units of endowment at t=1, s=1, and 4 units of endowment at t=1, s=2) and the endowment for agent B is $e^B = (5, (4, 2))$ (3 units of endowment at t=0, 4 units of endowment at t=1, s=1, and 2 units of endowment at t=1, s=2). Both agents' Bernoulli utility, i.e. small u, from consumption follow the logarithm form log(c). The discount factor for future consumption is $\beta$=0.9. We assume that it is possible to trade a set of Arrow Debreu securities that pay off one unit of consumption good if and only if a given state of nature occurs (two securities in total). As in class, we let $\mathbf{q_1}$ denote the price of an Arrow-Debreu security that pays off in state 1 and let $\mathbf{q_2}$ denote the price of the Arrow-Debreu security that pays off in state 2, the price of current consumption good is 1. We are now set to determine equilibrium allocation and prices for this economy.

Note: when the questions below only ask you to formulate some problem, you just need to list related equations/functions, and don't need to solve the equations.

**Q6. A [6 points]**

Formulate the agent A investment problem given the prices of state contingent claims and express the demand for each state contingent claim, i.e. consumption, as a function of the prices $\mathbf{q_1}$ and $\mathbf{q_2}$.

**Q6. B [6 points]**

List the market clearing equations for all the periods and states suppose that there are 1/2 of type A investors and 1/2 of type two investors in the economy as listed in the question.

**Q6. C [6 points]**

Now suppose instead that there are 2/3 of type A investors and 1/3 of type two investors in the economy.

Q6.C.1. Is there change to the answer of Q6.A, i.e., agent A's consumption given the prices of state contingent claims? If yes, how? If no, no further explanation. **[3 points]**

Q6.C.2. List the market clearing equations for all the periods and states **[3 points]**.

**Q6. D [6 points]** Let's go back to assume that there are 1/2 of type A investors and 1/2 of type B investors, is it possible that the equilibrium allocation of consumption is $c^A = (c_0^A, (c_1^A, c_2^A)) = (3, (3.5, 4.5))$? Why or why not?

**Q6. E [6 points]** Let's assume that there are only two agents in the economy, one of which is a type A investor and the other is a type B investor, each with the same endowments as mentioned above. Now, let's imagine that there is a social planner whose utility function puts 1/3 of weight on investor A's utility while 2/3 of weight on investor B's utility in allocating consumption goods to the two investors. Formulate the social planner's problem in allocation consumptions to investor A and B.

**Q6. F [5 points]** Continue with the social planner's set up in question Q6.E, is it possible that $c^A = (c_0^A, (c_1^A, c_2^A)) = (2, (1.5, 2.5))$ being a pareto optimal solution? Why or why not?

**Question 7 [10]**

Stock in company Lu is currently selling for $200 per share, and over the next period will either go up by 40% (probability of occurrence = 0.6) or go down by 40% (probability of occurrence = 0.4). The interest rate over the period is 5% and the stock pays no dividends over the period. Consider a call option on company Lu stock with a striking price of $100 which expires at the end of the period.

**Q7.A [3 points]** What is the current value of the call option?

**Q7.B [3 points]** Suppose instead the stock price can either rise by 40% or fall by 80%, other things equal. What is the value of the call option then?

**Q7.C [4 points]** Continue with the scenario in Question 7.B. Calculate the expected payoff of the option, and explain intuitively why the expected payoff of the option is not the same as the current market value of the call option?